Cognitive Companion — team and engineering backing

A briefing document for department leadership. Clinical home in Anesthesiology. Engineering partner already engaged at the College of Engineering.

Why this is not a solo side project

Cognitive Companion fails if it is only a clinician with a slide deck, and it fails if it is only a computer-science prototype that has never been in a sim bay. Year one is a joint build: anesthesiology owns the problem, the scenarios, and the measurement; software engineering owns implementation quality so we do not ship a toy.

Clinical team

Sergio Navarrete, D.O. — transplant anesthesiology and critical care, VCU. Clinical lead and product owner for version 1. The work is grounded in transplant ICU cognitive load and in simulation as the ethical place to test packaging before any live-patient claim.

Brian Kazior, M.D., and Lorena Beltoja, M.D. — clinical collaborators. They are part of the intended sim-facing team for scenario judgment and cue review. They are not being asked to fund this. They are being asked to keep the content honest.

Version 1 does not put this device on live transplant patients. The clinical gravity of that unit is why the training problem is worth solving. The simulation lab is where we are allowed to measure misses.

Engineering partner we already have

We have already spoken with **Rodrigo Spínola, Ph.D.**, Associate Professor of Computer Science at the VCU College of Engineering and Co-Director of the Software Engineering Center. His research is empirical software engineering, software quality, and technical debt. He is the principal investigator of a current NSF award building “living labs” so computer-science students learn to develop, deploy, and service AI systems for health.

Dr. Spínola has reviewed this project. He believes it has a substantial likelihood of true implementation success. That sentence is the engineering backing we are bringing to the chair. It is not a letter of support we are inventing after the fact. It is a conversation that already happened.

What that backing is **not**: it is not a funded subcontract. It is not NSF money reallocated to this project. It is not a claim that Computer Science will staff year one for free. It is an expert implementation judgment from the person we would actually build with.

Why Spínola is the right partner, not a cold call

In April 2025 the National Science Foundation awarded $750,000 to Dr. Spínola for AIHealth-LL, an experiential living-lab program against the shortage of AI professionals in healthcare. Co-principal investigators are **Kostadin Damevski, Ph.D.**, Computer Science, and **Daniel Falcão, D.O., FAAN**, Associate Professor of Neurology at VCU and Chief of the Division of Vascular Neurology and Critical Care.

That program is already a College of Engineering plus School of Medicine collaboration. Undergraduate computer-science students work with neurology and VCU Health on developing and servicing AI systems for health.

New coursework includes Artificial Intelligence Applied to Health and Engineering AI Systems for Health. Partners named on that award include the VCU Institute for Engineering and Medicine and the Department of Neurology.

Dr. Falcão is a neurologist and neurocritical-care leader working with Dr. Spínola on multimodal artificial intelligence in medical applications. That ongoing work is why we can say the engineering relationship is not hypothetical. We are not asking Anesthesiology to invent a campus partnership. We are asking Anesthesiology to seed a project that the software-engineering faculty who already do AI-for-health with clinicians have reviewed and judged implementable.

We are **not** claiming Dr. Falcão has reviewed Cognitive Companion. We are **not** claiming this project is an aim of the NSF award. We are claiming adjacency and capability: the people who know how to engineer AI-for-health at VCU have looked at this, and the principal engineer said it can be built.

How we will use that partnership in year one

Year-one engineering is the majority of the real cost. The seed should buy a start of that work under software-engineering standards — requirements that do not rot, a protocol-pack format that can be audited, a safety gate that is a test rather than a hope, and students or research staff who already sit in a lab that takes healthcare AI seriously.

Clinical faculty will not write production software between cases. Computer-science faculty will not invent malignant-hyperthermia steps. The collaboration is the product plan.

Governance we will keep boring on purpose

- Clinical owner: Navarrete, with Kazior and Beltoja on content.
- Engineering owner: Spínola’s software-engineering practice, not an anonymous contractor and not a vendor SDK we cannot inspect.
- Privacy owner: camera-free as a hard product rule, not a setting.
- Measurement owner: simulation faculty and the clinical team, process endpoints, no outcome theater.
- Money owner: Anesthesiology seed as springboard; School of Medicine research and other internal sources for the rest of year one; industry last because of conflict of interest.

If those five stay named, this does not become a pair of glasses in a drawer.